

Gait and Movement

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Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

- Objective 1: Explain how to take a proper history of a patient with Parkinson's
- Objective 2: Describe how to observe the patient (LOOK)
- Objective 3: Compare and contrast symptoms with increased or decreased muscle tone- passive exam (FEEL).
- Learning Objective 4: Assess strength- active exam (MOVE)
- Learning Objective 5: Observe the gait.

History Taking

- Age
- Background (some conditions are seen in certain groups)
- Age at onset (how many years)
- Handedness
- Symptoms at onset
- Prodromal Symptoms

Medical History

- What other medical problems exist?
- Psych history- anti psychotic use in the past
- Cardiac? Diabetes? Cancer? GERD?
- Was an anti-emetic used long term?
- This is also important in terms of medication use

Goal: rule out secondary/drug-induced parkinsonism

Current and past medication use

- What medications are currently being used?
- Certain medications- steroids, valproic acid, lithium may cause tremor.
- Dopamine blockers → parkinsonism
- Levodopa → on/off periods
- Dopamine agonists → impulse control, sleep attacks
- Remember to make a note of allergies to any medications.

Family History

- Does anyone have PD in the family?
- If so, at what age did the symptoms begin?
- Did anyone have a tremor?
- Use a cane, walker or wheelchair to get around?

Most PD is sporadic; however, family history raises suspicion for genetic forms.

Genetics of PD

- 15% of patients have family history
- Individuals of Ashkenazi Jewish descent are at a higher risk
- Most common genes: LRRK2, GBA1, PRKN
- Less common genes: SNCA, PINK1, PARK7, VPS35
- Core mechanisms: alpha-synuclein aggregation, mitochondrial dysfunction, impaired protein degradation
- Genetic Testing considered when: age of onset <50 and/or strong family history
- Even in genetic form of PD, environmental exposures strongly influence when and whether the disease develops.

Exposures

- Military service (i.e. agent orange in Vietnam/camp lejeune)
- Live/work near farm (sprayed pesticides)
- Vineyard exposure (sprayed pesticides)
- Pesticides include rotenone, paraquat, trichloroethylene, glyphosate (Roundup)
- Head trauma (sports injuries or MVA)
- Live/work near a golf club
- Live/work near a dry cleaners (Perchloroethylene, PERC)
- Rural Living- well water
- Drugs- MPTP

Psychiatric History

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- Did the patient ever have a history of psychosis that was treated with medications?
- Medications may include anti-dopaminergics (haloperidol) or other anti-psychotics.

Past surgeries

- In addition to asking about surgeries- observe for scars.
- Do not miss absent extremities (fingers, toes, legs, etc.)

Sleep

- Ask about vivid dreams
- Injuring bed partner or self (REM sleep behavior disorder)
- Restless leg syndrome

Can precede motor symptoms by years

Review of Systems

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- Loss of smell (when first noticed)
- Constitutional Symptoms- weight loss common
- GI (constipation)
- Cardiac (lightheaded)
- Pulmonary (SOB)- sometimes
- GU- frequent urination, especially at night
- Derm- Seborrheic dermatitis

Very Important

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- Any choking or coughing on foods or liquids?
- Any falls?

Think atypical parkinsonism

- Mental status- assess for speech quality- clarity and strength?
- Good historian?
- If concerns- then mental status examination (MOCA best)

Cranial Nerves

- Vertical gaze limitation → consider PSP
- Do not forget to check for shoulder lag.
- Screen for sleep apnea
- Hypomimia (masked facies)
- Decreased blink rate

Motor Exam

- In addition to strength, check the tone.
- Lead-pipe rigidity
- Cogwheeling
- Do not forget to compare rigidity from side to side.
- Check for rapid alternating movements
- Diagnosis requires bradykinesia with rigidity (+/- tremor)

Tremor, sensory, reflexes

- If tremor:
- Resting, action, postural?
- Resting → Parkinson's
- Postural/action → essential tremor
- Is there also a neuropathy that may account for falls?
- Check reflexes: including plantar and Hoffmann

Gait Exam

- Especially look for arm swing
- Wide or narrow based gait
- Weight bearing
- Sacral base unleveling (one hip lower than the other).
- Shuffling steps
- Stooped posture
- En bloc turning
- Freezing

Parkinson Diagnostic Tests

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- Clinical diagnosis- physical and neurological examination
- Dopamine transporter (DAT) scan
- Genetic testing- useful if there is known family history or have early-onset disease
- Levodopa challenge- improvement in motor symptoms following dopaminergic therapy
- Alpha-synuclein test- clumps of protein alpha-synuclein can be obtained through a skin biopsy or CSF lumbar puncture; researchers hope the test could be done using blood samples in the future

Whole-body condition – "S" words

- Substantia nigra
- Synucleinopathy
- Shake
- Stiffness
- Sialorrhea
- Stare
- See things (hallucinations)
- Stooped
- Shuffle
- Syncope
- Sleep disorder
- Swallowing dysfunction
- Sleepy
- Sexual disorder
- Scared
- Sad
- Smell (anosmia)
- Sweating
- Sniffle
- Seborrheic dermatitis
- Shoulder pain
- Slurred speech
- Socially isolated
- Slowed cognition
- Slow colonic transit
- Storage urinary symptoms

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Practice Question 1

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A 62-year-old man presents with progressive difficulty walking and slowness of movement. On physical exam, he has decreased facial expression, reduced left arm swing, cogwheel rigidity of the left upper extremity, and decreased blink rate. He also reports loss of smell for several years as well as vivid dreams which he acts out.

Which finding is **most essential** to establish the diagnosis of Parkinson's disease in this patient?

- A. Loss of smell
- B. Masked facies
- C. Bradykinesia
- D. Resting tremor
- E. REM sleep behavior disorder

Practice Question 1

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A 61-year-old man presents with progressive difficulty walking and slowness of movement. On physical exam, he has decreased facial expression, reduced left arm swing, cogwheel rigidity of the left upper extremity, and decreased blink rate. Patient has left resting tremor as well. He also reports loss of smell for several years as well as vivid dreams which he acts out. Which finding is **most essential** to establish the diagnosis of Parkinson's disease in this patient?

- A. Loss of smell
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Explanation: Parkinson's disease requires bradykinesia for diagnosis. Resting tremor is not necessary for diagnosis of PD. Anosmia and REM sleep behavior disorder are both prodromal features, but they do not establish the diagnosis.

Practice Question 2

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A 71-year-old male patient is observed walking into the clinic. He has a stooped posture, reduced arm swing, shuffling steps, and turns around using multiple small steps. When asked to start walking, he hesitates and feels as though his feet are “stuck to the floor.” Which underlying neurologic dysfunction best explains this disease?

- A. Corticospinal tract lesion
- B. Cerebellar hemispheric dysfunction
- C. Basal ganglia dysfunction
- D. Dorsal column dysfunction
- E. Vestibular system dysfunction

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- A. Corticospinal tract lesion--> spasticity
- B. Cerebellar hemispheric dysfunction-->wide-based, ataxic gait
- C. Basal ganglia dysfunction
- D. Dorsal column dysfunction--> sensory ataxia, positive Romberg
- E. Vestibular system dysfunction--> imbalance, vertigo

Explanation: Shuffling, freezing, en bloc turning, and difficulty starting movement points to a dysfunction in the basal ganglia, since it controls voluntary movements and motor initiation.

Summary Slide

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- Review the importance of eliciting a solid HPI.
- Don't forget meds.
- Perform a careful examination.

THANK YOU!

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